Technical details of statistical analysis used in paper

Shifting Dimensions: The Transformation of Reproduction Sizes in Art Magazines (1900–1950)

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This document contains further details on statistical analysis that did not make it to the main manuscript because of the restricted length.

Modelling temporal trends

To fit temporal models, we used Bayesian statistics with unbiased permissive priors.

Four alternative models were estimated: one without a temporal effect, one including a linear time trend, one allowing for gradual non-linear change over time using basis spline functions, and finally an equally complex model of abrupt changes.

The relative image area (constrained between 0 and 1) was transformed to the logit scale before the analysis to allow for modelling the output with the normal distribution. This approach was also used to model relative image width. The logit ("log-odds") function takes a constrained measure such as ratio or probability p and maps it onto the whole real number axis by taking $\log(p/(1-p))$. Absolute image width (in principle not bounded from above) was modelled on the original scale.

Both models with the nonlinear effect of time, splines and steps, were characterized by 7 parameters in total (to not favor any of the models by randomly assigning it with complexity closer to an optimum). In the spline model, this meant using cubic splines with 5 internal knots resulting in 7 spline basis functions in total. In the step model, 7 parameters corresponded to 4 independent baselines, one for each interval, and 3 cutpoints between these intervals. Referential models with linear trend and with no effect of time had less parameters.

The complete table of WAIC results can be found in Table 1.

Table 1. Model comparison based on the expected out-of-sample fit

Relative image area (logit)				**Relative width (logit)**				**Absolute width**			
Model	WAIC	SE	weight	Model	WAIC	SE	weight	Model	WAIC	SE	weight
step	109720	329.48	0.79	linear	104622	317.344	0.46	step	326792	241.201	0.46
no_time	109723	329.49	0.15	step	104623	317.424	0.28	linear	326793	241.161	0.33
spline	109725	329.54	0.05	no_time	104625	317.459	0.15	no_time	326795	241.159	0.12
linear	109728	329.53	0.01	spline	104625	317.416	0.11	spline	326795	241.191	0.09

WAIC = Widely Applicable Information Criterion, SE = standard error, weight = relative model weight calculated as $\exp(-0.5\times dWAIC)/\Sigma\exp(-0.5\times dWAIC)$

Cluster analysis of image size: Discovering size regimes in each magazine

For each magazine at each timestep, we identified local modes (peaks in the density plot) in the distribution of image sizes. The resulting clusters were then linked across neighbouring time windows to form temporal tracks. These tracks allowed us to identify recurring size regimes and how they changed over time.

Each temporal window covered 3 years and was moved along the time axis in half-year steps. Within each window, the distribution of image size was estimated using kernel density estimation, with the bandwidth set to 0.7× the default value of Silverman's rule-of-thumb bandwidth selector. Local maxima of the resulting density function were treated as empirical modes of image size clusters; weak local maxima below 10% of the height of the strongest mode in the same window were discarded. A mode was assigned to an existing track when its distance from the nearest mode in the preceding window did not exceed 0.5 units on the logit-transformed area scale; otherwise, a new track was started. Tracks appearing in fewer than three temporal windows were removed.

To assess the sensitivity of this exploratory procedure to the selected algorithm parameters, we repeated the temporal-mode analysis with alternative settings for the window length, density bandwidth, local-mode threshold, maximum linking distance, and minimum track length. The overall interpretation of the results was robust to these changes: although individual short-lived tracks and minor local modes varied across parameter settings, the main distinction between changes in cluster use, within-cluster shifts, and cluster turnover remained stable.

Cross-entropy calculation

To assess convergence between magazines, we compared how reproduction size distributions differed between simultaneously published magazines within each temporal window.

We expressed the relative usage of different image sizes in each magazine at given time with a histogram and calculated the difference between every possible pair of magazine histograms from that time using a measure from information theory called Jensen–Shannon divergence ("How badly we would do if we predicted distribution of images from magazine A using distribution from magazine B and vice versa.")

The logit-transformed area scale was divided into 15 common bins, and for each magazine and time window we calculated the proportion of reproductions falling into each bin. We then calculated pairwise Jensen–Shannon divergences between magazines' binned size distributions and summarized these distances by their mean value in each temporal window. Jensen–Shannon divergence was used because it provides a symmetric and finite measure of dissimilarity between probability distributions. A decline in mean pairwise divergence indicates that magazines became more similar in their size composition, while stable or increasing divergence suggests persistent or growing differentiation between magazines.
Again, the analysis results proved to be robust to parameter value choices (number of histogram bins.). To ensure that the relative stability of yearly cross-magazine diversity does not mask abrupt changes between individual years, we also calculated Divergence from the distribution of magazines published in the previous year and averaged those yearly to get an alternative measure of relative distribution overturn. The two measures of magazine diversity were, however, very tightly correlated (r=0.67 in relative area, 0.75 in relative width, and 0.72 in

absolute width). This shows that no important change happens at the seam of neighbouring years. For this reason, we show only yearly cross-magazine diversity in the main article's results.